

Understanding Credit Card Fraud Through Data

A plain-language walkthrough of a fraud risk analysis project

What This Project Is About

Credit card companies lose money two ways: when they miss real fraud, and when they wrongly flag a normal purchase and annoy an honest customer. This project looks at 50,000 sample transactions to answer a simple question:

“Can we spot the warning signs of fraud before it happens, using patterns in the data?”

Note: this uses a made-up (synthetic) dataset built to behave like real transaction data, not actual customer information. This is standard practice for a portfolio project.

Step 1: How Rare Is Fraud, Really?

The first thing to check is: out of all transactions, how many are actually fraud? This matters because if fraud is very rare, a lazy prediction of “never fraud” would look accurate on paper while being completely useless.



Figure 1 — Count of fraud (1) vs. legitimate (0) transactions

What it shows: Out of 50,000 transactions, only a small sliver (around 800) were fraud. The vast majority — about 98% — were completely normal.

Why it matters: This confirms fraud is a rare-event problem. Any solution needs to be specifically built to catch rare events, not just “mostly right.”

Step 2: Where Does Fraud Happen Most?

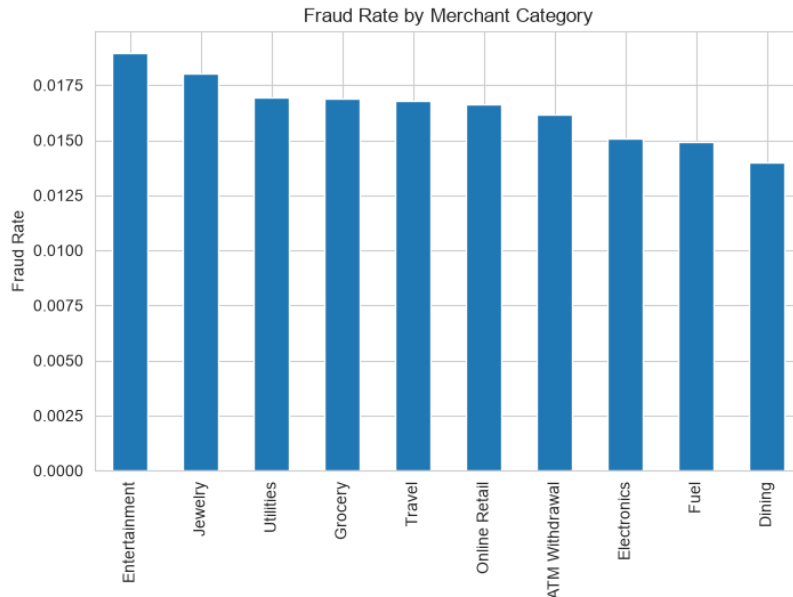


Figure 2 — Fraud rate by type of purchase

What it shows: Each bar is a category of purchase (Entertainment, Jewelry, Groceries, etc.), and the height shows what percentage of purchases in that category turned out to be fraud.

Why it matters: Entertainment and Jewelry purchases have the highest fraud rates. This tells a business exactly where to focus extra caution — for example, asking for extra verification specifically on big-ticket jewelry purchases, rather than slowing down every single transaction.

Step 3: Does Time of Day Matter?

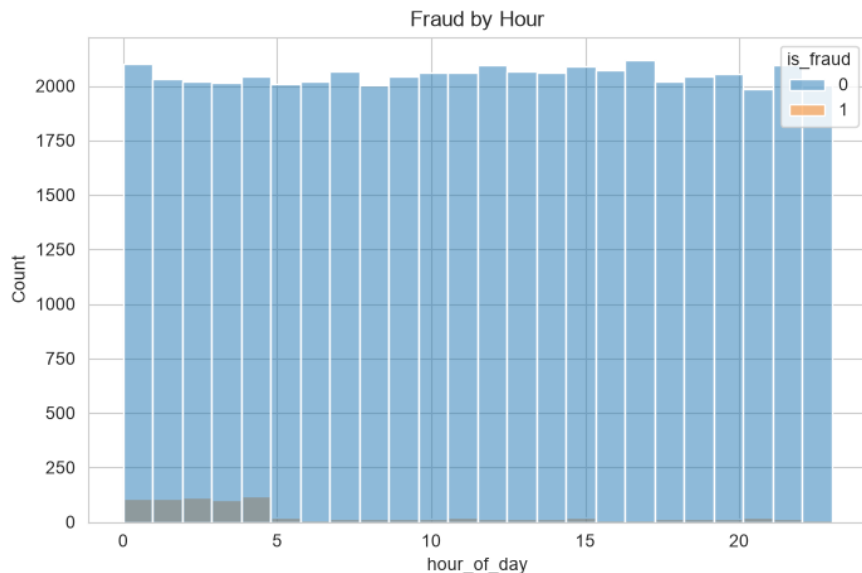


Figure 3 — Transaction volume by hour of day, fraud vs. legitimate

What it shows: Each bar is an hour of the day (0 = midnight, 23 = 11pm). The blue portion is normal transactions; the small orange sliver on top is fraud.

Why it matters: Late-night hours show a visibly higher share of fraud relative to overall volume, even though overall transaction volume is fairly flat across the day. This is a useful early-warning signal: transactions made in the middle of the night deserve a slightly closer look.

Step 4: Does Distance From Home Matter?

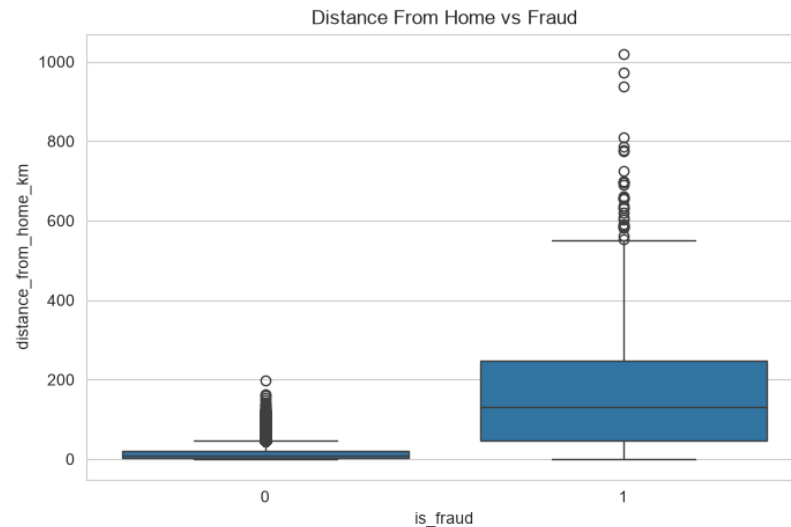


Figure 4 — How far the purchase was from the cardholder's home

What it shows: This is a box plot — think of it as a visual summary of “typical range.” The box on the left (labeled 0) is legitimate purchases; the box on the right (labeled 1) is fraud. The higher the box sits, the further from home those purchases happened.

Why it matters: Legitimate purchases almost always happen close to home. Fraudulent purchases are typically much farther away — often 100+ km. This is one of the strongest, most intuitive red flags: a purchase suddenly happening far from where someone lives is worth a second look.

Step 5: Which Factors Are Most Connected to Fraud?

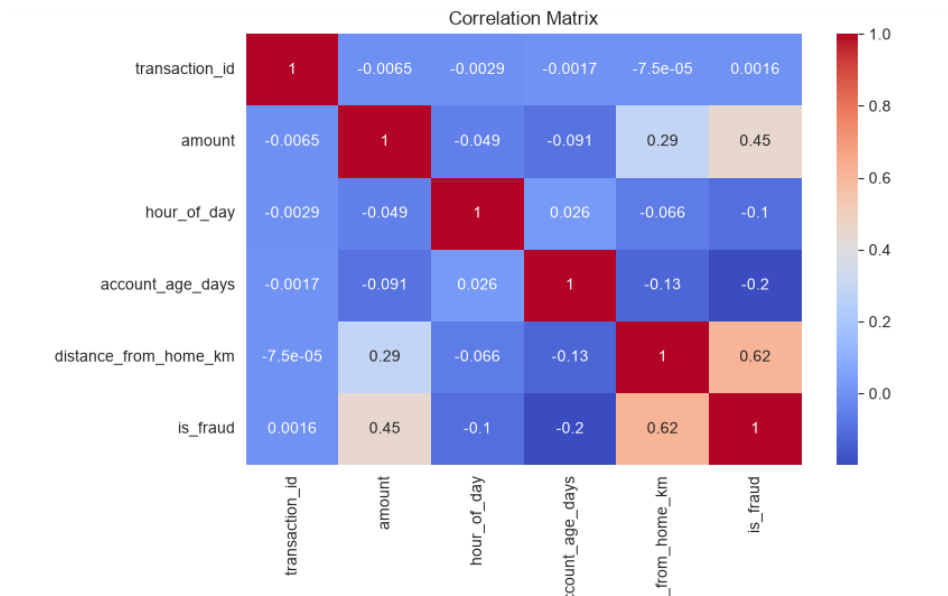


Figure 5 — Correlation matrix: how strongly each factor relates to every other factor

What it shows: This grid compares every factor in the data against every other factor. Dark red means “strongly connected,” dark blue means “strongly opposite,” and pale colors mean “not very related.” Look at the bottom row (“is_fraud”) to see what's most connected to fraud.

Why it matters: Distance from home (0.62) and purchase amount (0.45) stand out as the two factors most strongly linked to fraud. Account age is negatively linked (-0.2), meaning newer accounts are somewhat more associated with fraud. This step is how you decide which clues are actually worth building a detection system around, instead of guessing.

Step 6: Building a Prediction Model

With the important warning signs identified, the next step is to build a model — a computer program that looks at a new transaction and estimates: “how likely is this to be fraud?” The type of model used here is called Logistic Regression, a well-established, easy-to-explain method (as opposed to a “black box” model nobody can interpret).

LogisticRegression ⓘ ?		
▼ Parameters		
penalty		'deprecated'
C		1.0
l1_ratio		0.0
dual		False
tol		0.0001
fit_intercept		True
intercept_scaling		1
class_weight		None
random_state		None
solver		'lbfgs'
max_iter		100
verbose		0
warm_start		False
n_jobs		None

Figure 6 — The model's configuration settings

What it shows: This is simply the model's settings — think of it like the settings menu of an app. The important one here is a setting called “class_weight,” which controls how much the model prioritizes catching rare fraud cases.

▼ Fitted attributes		
Name	Type	Value
classes	ndarray[int64](2,)	[0,1]
coef	ndarray[float64](1, 13)	[[2.22,-0.82,-5.2 , ..., -0.16,-0.06,-0.08]]
intercept	ndarray[float64](1,)	[-12.7]
n_features_in	int	13
n_iter	ndarray[int32](1,)	[15]

Figure 7 — What the model learned from the data

What it shows: After being shown thousands of past transactions, the model produces a set of internal numbers (“coefficients”) representing how much weight it gives each factor. This is the model's “learned knowledge.”

Step 7: How Good Is the Model, Actually?

The real test: give the model transactions it has never seen before, and check how often it gets the answer right.

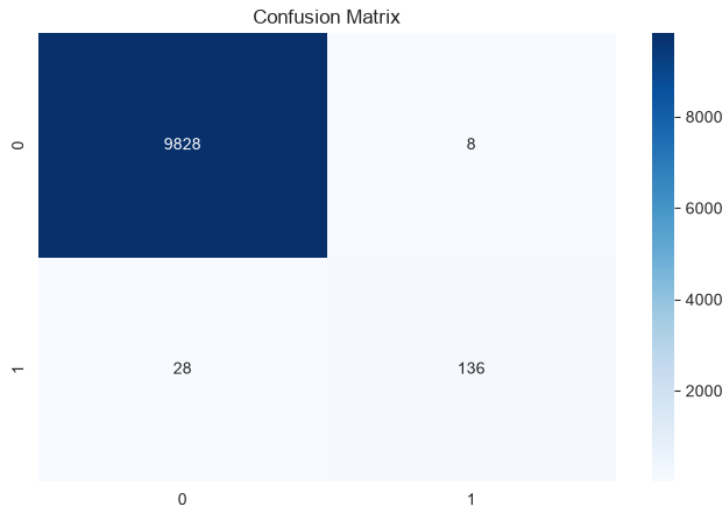


Figure 8 — Confusion matrix: predicted vs. actual results

What it shows: This table compares what the model predicted against what actually happened, on transactions it hadn't seen during training.

	Model said: Legitimate	Model said: Fraud
Actually Legitimate	9,828 correct	8 wrongly flagged
Actually Fraud	28 missed	136 correctly caught

Plain-English translation:

- Out of 164 real fraud cases, the model correctly caught 136 of them — about 83%.
- It only wrongly flagged 8 normal purchases out of 9,836 — a very low false-alarm rate.
- It missed 28 fraud cases — the main area where the model could still improve.

What This Means for the Business

- Distance from home and purchase amount are the two strongest fraud signals — worth building real-time alerts around.
- Late-night transactions and Entertainment/Jewelry purchases carry extra risk and could warrant additional verification steps.
- The model catches most fraud (83%) while rarely bothering honest customers (99.9% of legitimate purchases went through smoothly).
- No model is perfect — the 28 missed cases show there's room to add more signals (like how fast someone is making repeat purchases) in a future version.

Why This Approach, Not Just “High Accuracy”

A simple but important idea in this project: if 98% of transactions are legitimate, a model that always guesses “not fraud” would be 98% “accurate” while being completely useless — it would catch zero fraud. That's why the real measures used here are how many actual fraud cases were caught (recall) and how many alerts were correct (precision), not overall accuracy. This distinction is exactly the kind of judgment a business analyst is expected to bring — not just running a model, but knowing which numbers actually matter for the business decision at hand.